ELABORATOR

Survey Results Report

Sustainable Urban Mobility Application (SMARTA App)

Trikala Pilot

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UKRI supports UK participants with grant numbers 10078745 (iRAP) and 10069203 (UBRIS).

Co-funded by the European Union

UK Research and Innovation

Document Control Page

Deliverable title	Survey Results Report_ Sustainable Urban Mobility Application (SMARTA APP)_ Trikala Pilot
Due date of delivery	29/04/2026
Dissemination level	[Internal, Public, Restricted / European Commission]
Type	[Report,]
Editor(s)	[Christina Karamperi, e-trikala]
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Submitted by (Project Coordinator)	[Name, organisation]
Project name	The European Living Lab On Designing Sustainable Urban Mobility Towards Climate Neutral Cities
Project Acronym	ELABORATOR
Project starting date	01/06/2023
Project duration	42 months
Rights	ELABORATOR consortium

Document History

Version	Date	Beneficiary	Description
0.1	DD/MM/YYYY	[Name, organisation]	
0.2	*DD/MM/YYYY*	*[Name, organisation]*	
0.3	*DD/MM/YYYY*	*[Name, organisation]*	

Project Executive Summary

ELABORATOR stands for 'The European Living Lab on designing sustainable urban mobility towards climate neutral cities'. The EU-funded project uses a holistic approach for planning, designing, implementing and deploying specific innovations and interventions towards safe, inclusive and sustainable urban mobility. These interventions consist of smart enforcement tools, space redesign and dynamic allocation, shared services, and integration of active and green modes of transportation.

They will be specifically co-designed and co-created with identified "vulnerable to exclusion" user groups, local authorities and relevant stakeholders. The interventions will be demonstrated in a number of cities across Europe, starting with six Lighthouse cities and six Follower cities with three principal aims:

I. to collect, assess and analyse user needs and requirements towards a safe and inclusive mobility and climate neutral cities;
II. to collect and share rich information sets made of real data, traces from dedicated toolkits, users' and stakeholders' opinions among the cities, so as to increase the take up of the innovations via a twinning approach;
III. to generate detailed guidelines, policies, future roadmap and built capacity for service providers, planning authorities and urban designers for the optimum integration of such inclusive and safe mobility interventions into Sustainable Urban Mobility Plans (SUMPs).

ELABORATOR Lighthouse cities

- Milan (Italy)
- Copenhagen (Denmark)
- Helsinki (Finland)
- Issy-les-Moulineaux (France)
- Zaragoza (Spain)
- Trikala (Greece)

ELABORATOR Follower cities

- Lund (Sweden)
- Liberec (Czech Republic)
- Velenje (Slovenia)
- Split (Croatia)
- Krusevac (Serbia)
- Ioannina (Greece)

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List of abbreviations and acronyms

Acronym	Meaning
ASAP	As Soon As Possible
B2B	Business-to-business
B2C	Business-to-Consumer
EC	European Commission
GA	Grant Agreement
KoM	Kick-off Meeting
KPI	Key Performance Indicator
WP	Work Package
MS	Milestone

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Deliverable executive summary

This report presents the findings of a user survey assessing the performance, usability, and inclusiveness of a sustainable mobility application implemented in Trikala. The survey gathered responses from 118 participants. Overall, the application demonstrates satisfactory usability and functionality, while improvements are required in inclusiveness and accessibility, particularly for vulnerable groups and migrants.

1. Methodology

The survey was administered through an online questionnaire using a five-point Likert scale (1–5). It assessed user perceptions across key dimensions, including usability, functionality, mobility services, and inclusiveness. The collected data were analysed using descriptive statistics (mean values) and presented through bar charts (see Annex).

The survey was disseminated via e-Trikala's social media channels and remained open for one month (February 2026). In total, 118 participants responded. The findings were subsequently discussed and validated during a stakeholders' workshop held on 8 April 2026 at the Local Innovation Hub.

Figure 1 – Social Media Poster with the survey QR

Figure 2- Stakeholders meeting and discussion_ 8th of April 2026

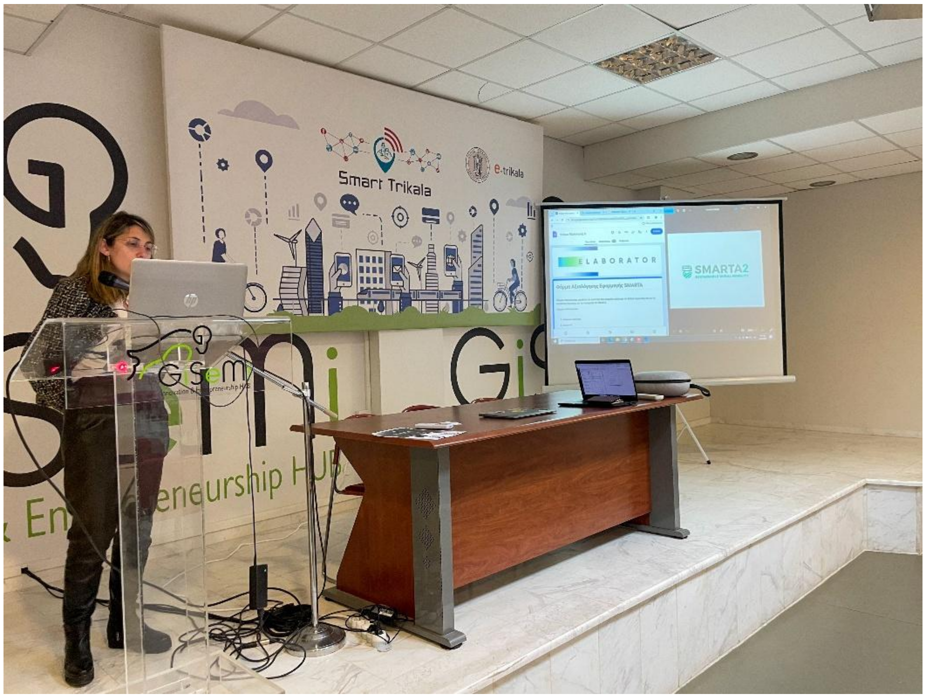

Figure 3- Stakeholders meeting and discussion_ 8th of April 2026

2. Results Analysis

The analysis of the survey responses provides a comprehensive overview of user perceptions regarding the SMARTA mobility application in Trikala. A total of 118 participants contributed to the survey, offering insights across four key dimensions: usability, functionality, mobility services, and inclusiveness.

2.1Overall Performance

Overall, the application received positive evaluations, with most responses clustering between moderate and high satisfaction levels (scores 3 to 5). This indicates that users generally perceive the application as a useful and functional tool for supporting urban mobility. However, variability across specific features suggests areas requiring targeted improvements.

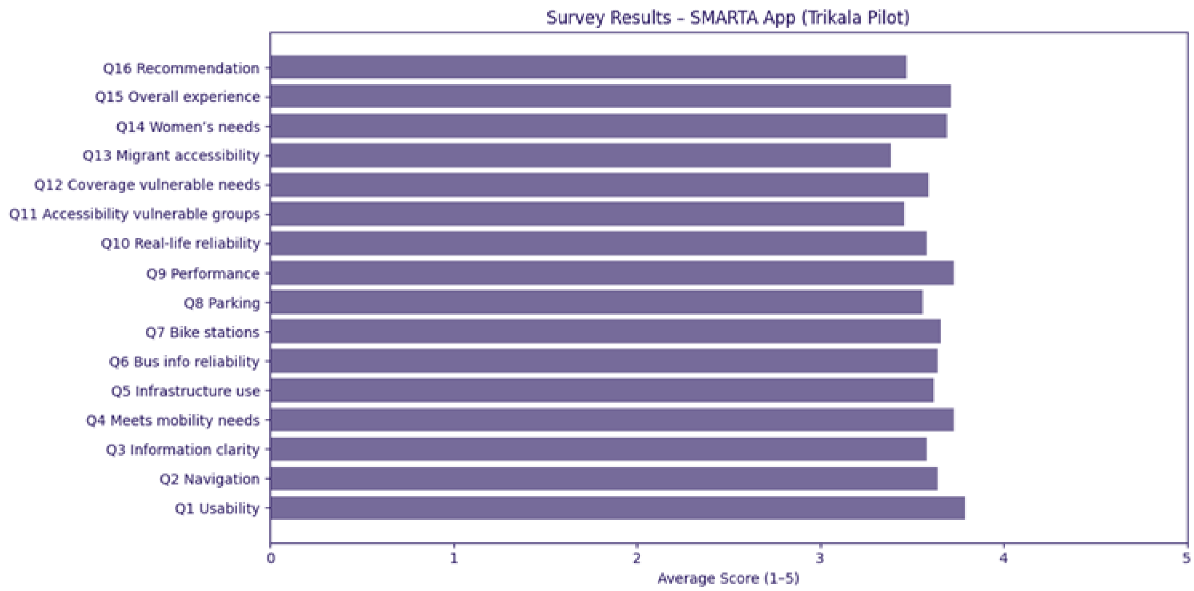

Figure.4 – Overall performance results

Question	Average Scores	Summary
Q1. The application is well organized and user-friendly	3.79	Users perceive the app as generally well-structured and easy to use.
Q2. Navigation is simple and understandable	3.64	Navigation is satisfactory, though some improvements are needed.
Q3. Information provided is clear and informative	3.58	Information clarity is acceptable but not optimal.
Q4. The app meets mobility needs of Trikala residents	3.73	The app generally responds well to user mobility needs

Q5. Effective use of sustainable mobility infrastructure	3.62	Good integration with city infrastructure, with room for improvement.
Q6. Reliable bus routes and stops information	3.64	Bus-related information is considered fairly reliable.
Q7. Helps locate shared bike stations	3.66	Users find bike station location features useful.
Q8. Helps locate parking spaces	3.56	Parking functionality is weaker compared to other features.
Q9. App performance (speed & technical reliability)	3.73	Performance is good with limited technical issues.
Q10. Reliability in real-life conditions	3.58	Moderate reliability during actual use scenarios.
Q11. Accessibility for vulnerable groups	3.46	Accessibility is relatively weak and needs improvement.
Q12. Coverage of vulnerable groups' mobility needs	3.59	Partial coverage of needs, but not fully sufficient
Q13. Accessibility for migrants (e.g. language support)	3.39	This is the weakest aspect—limited inclusiveness for migrants.
Q14. Consideration of women's mobility needs	3.69	Fairly positive perception regarding gender-related needs.
Q15. Overall user experience	3.71	Overall experience is positive.
Q16. Willingness to recommend the app	3.47	Users are moderately willing to recommend the app.

Table.1 – Detailed Results per question

2.2 Usability

The usability dimension scored consistently well, reflecting that users find the application relatively easy to use and navigate. Key aspects such as interface clarity, ease of access to core functions, and general user experience were positively rated.

Despite this, some respondents indicated challenges in navigation and information structure, suggesting that further refinement of the user interface (UI) and user experience (UX) design could enhance overall usability. Simplifying navigation pathways and improving visual hierarchy would likely improve user satisfaction.

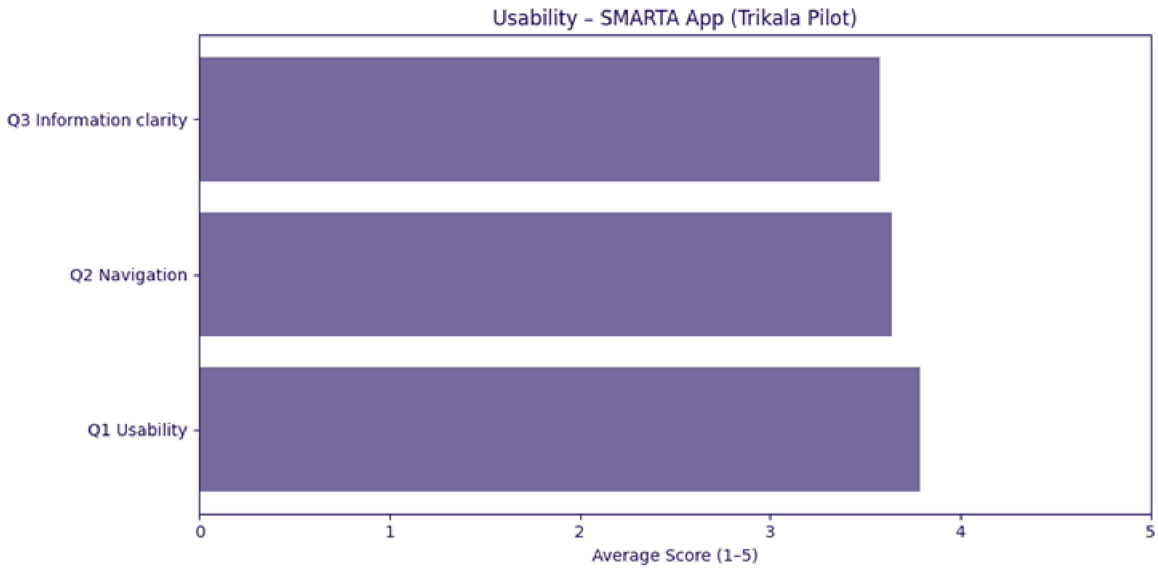

Figure.5 – Usability scores of Smarta

2.3 Functionality & Mobility Services

The functionality of the application was also rated positively, particularly in relation to its core services. Users acknowledged the availability of essential features supporting mobility needs, such as route planning and general information provision.

However, certain functionalities—especially those related to real-time data—received comparatively lower scores. This indicates a need to improve system reliability, data accuracy, and responsiveness to ensure a more seamless and dependable user experience.

User perceptions of mobility services were generally moderate. While the application provides access to relevant transport-related information, specific services—most notably parking-related features—received lower evaluations.

This suggests that users may experience limitations in accessing accurate, real-time, or comprehensive information regarding parking availability and related services. Enhancing these features could significantly improve the practical value of the application for daily mobility needs.

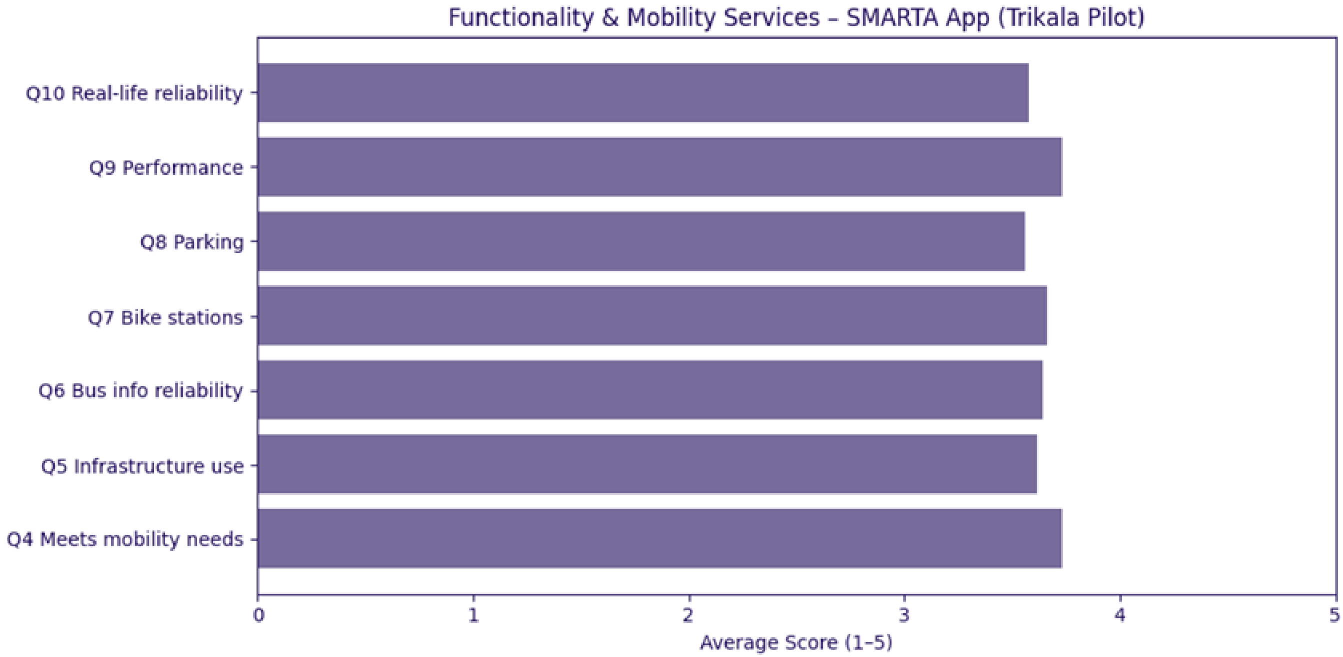

Figure. 6 – Functionality and Mobility Services of Smarta

2.4 Inclusiveness

Inclusiveness emerged as the weakest-performing dimension in the survey. Responses indicate that the application does not yet adequately address the needs of all user groups, particularly vulnerable populations and migrants.

Challenges identified include limited accessibility features, lack of multilingual support, and insufficient adaptation to diverse user needs. This highlights a critical area for improvement, especially considering the project's objective of promoting inclusive and equitable urban mobility solutions.

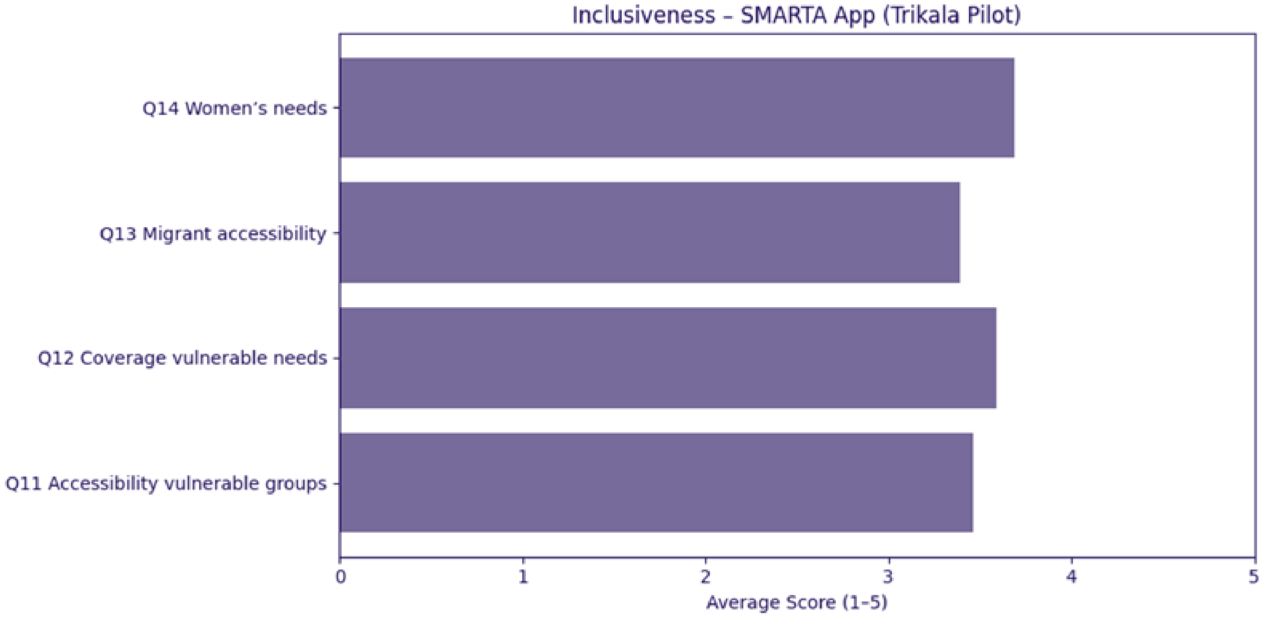

Figure. 7 – Inclusiveness and accessibility of Smarta

2.5 Variability of Responses

The distribution of responses across questions reveals some variability, indicating differing user experiences. While a significant proportion of users expressed satisfaction, a notable number provided neutral or lower ratings, particularly in areas related to inclusiveness and specific service functionalities.

This variation suggests that while the application performs well for a general audience, it may not yet fully meet the expectations or needs of all user segments.

2.6 Validation through Stakeholder Engagement

The survey results were further discussed during a stakeholder workshop held on 8 April 2026 at the Local Innovation Hub. The discussion confirmed the overall positive perception of the application, while also reinforcing the need to address identified gaps, particularly in inclusiveness and real-time service functionality.

Stakeholders emphasized the importance of continuous improvement through user feedback, co-creation processes, and iterative development to ensure that the application evolves in line with user needs and expectations.

3. Key Findings

The analysis of the survey results provides valuable insights into user perceptions of the SMARTA application and highlights both its strengths and areas requiring further development. Overall, the findings confirm that the application constitutes a robust digital tool supporting sustainable urban mobility in Trikala, while also revealing specific gaps that need to be addressed to fully meet user expectations and project objectives.

Firstly, the application demonstrates **consistently positive performance in terms of usability and functionality**, with average scores ranging between 3.6 and 3.8 across related indicators. Users generally perceive the application as intuitive, well-structured, and easy to navigate, confirming the effectiveness of its current design and core functionalities.

Secondly, the results indicate that **mobility-related services are adequately supported**, particularly in areas such as route information and access to shared mobility options (e.g. bike stations). However, specific services—most notably those related to **parking availability and real-time data provision**—received comparatively lower scores. This suggests that while the application is operationally effective, its added value could be significantly enhanced through improved data accuracy and real-time capabilities.

Thirdly, and most critically, the analysis highlights that **inclusiveness remains the weakest dimension** of the application. Lower scores related to accessibility for vulnerable groups and migrants (especially regarding language support) indicate that the application does not yet fully align with the inclusive mobility objectives of the ELABORATOR project. This gap is particularly important given the project's emphasis on co-creation with "vulnerable to exclusion" user groups.

In addition, the findings reveal a **moderate level of overall user satisfaction and willingness to recommend the application**, suggesting that while the user experience is generally positive, further improvements are necessary to increase user trust, engagement, and long-term adoption.

Finally, the variability observed in responses indicates that **user experience is not uniform across all user groups**, reinforcing the need for a more user-centred and adaptive approach in future development phases.

4. Recommendations

Based on the survey findings and the stakeholder validation process, a set of targeted recommendations is proposed to enhance the performance, inclusiveness, and overall impact of the SMARTA application. These recommendations are aligned with the broader objectives of the ELABORATOR project and support its replication and scalability potential across European cities.

4.1 Enhancing User Experience and Usability

- Further optimise the **user interface (UI) and user experience (UX)** to improve navigation clarity and reduce complexity.
- Simplify access to key functionalities through improved visual hierarchy and streamlined user flows.
- Introduce **user guidance features** (e.g. onboarding tutorials or contextual help) to support new users.

4.2 Strengthening Functionality and Data Reliability

- Improve the **accuracy, availability, and real-time nature of mobility data**, particularly for parking and dynamic transport information.
- Enhance system responsiveness and technical reliability to ensure a seamless user experience.
- Explore integration with additional data sources (e.g. municipal systems, IoT sensors) to enrich functionality.

4.3 Improving Inclusiveness and Accessibility

- Introduce **multilingual support**, prioritising languages relevant to migrant communities.
- Integrate **accessibility features** (e.g. simplified interfaces, voice support, adaptable content) to better serve vulnerable groups.
- Adopt a **co-creation approach with target user groups**, ensuring that future updates reflect diverse needs and perspectives.

4.4 Increasing User Engagement and Adoption

- Implement targeted **awareness campaigns and training activities** to promote the application among different user groups.
- Leverage local stakeholder networks (e.g. community centres, NGOs) to reach underrepresented populations.
- Encourage user feedback through in-app mechanisms and participatory processes.

4.5 Supporting Continuous Improvement and Replication

- Establish a **continuous monitoring and feedback loop**, enabling iterative improvements based on user data and KPIs.
- Align application development with **Sustainable Urban Mobility Plans (SUMPs)** and local policy frameworks.
- Document lessons learned and best practices to support **replicability and transferability** to other ELABORATOR cities.

5. Conclusions and next steps

The results of the online survey, combined with the insights gathered during the stakeholder engagement process, confirm that the SMARTA application constitutes a **solid and functional digital solution** supporting sustainable urban mobility in the Municipality of Trikala. The overall positive evaluation across usability and functionality dimensions demonstrates that the application has successfully established a **reliable baseline for user-oriented mobility services**.

At the same time, the analysis highlights specific areas requiring further development, most notably in terms of **inclusiveness, accessibility, and the provision of real-time mobility information**. These findings are particularly relevant in the context of the ELABORATOR project, which places strong emphasis on **co-creation, user-centred design, and the inclusion of vulnerable-to-exclusion groups**.

In this context, the transition towards the **SMARTA 3 integrated application** represents a critical next step for the Municipality of Trikala. As presented during the stakeholder workshops, SMARTA 3 introduces an enhanced and more comprehensive mobility ecosystem, incorporating advanced functionalities such as:

- improved public transport information and route optimisation,
- support for shared and on-demand mobility services,
- integration of multiple transport modes within a unified platform,
- enhanced user interaction and service personalisation features.

These functionalities, as illustrated in the SMARTA 3 technical description, aim to transform the application from a standalone information tool into a **fully integrated smart mobility platform**, aligned with the principles of Mobility as a Service (MaaS) and sustainable urban planning.

Importantly, during the stakeholder meetings held on 8 April 2026, the Municipality of Trikala formally presented the SMARTA 3 application and explicitly acknowledged the need to **take into serious consideration the findings of the online survey**. Stakeholders emphasised that the future deployment of SMARTA 3 should directly address the identified gaps, particularly:

- improving accessibility for vulnerable user groups,
- enhancing multilingual support for migrant populations,
- strengthening real-time data accuracy and service reliability,
- ensuring a more inclusive and user-friendly interface.

This iterative approach reflects a **continuous improvement cycle**, where user feedback and stakeholder input are actively integrated into the development process, in line with Living Lab methodologies promoted by ELABORATOR.

In conclusion, the SMARTA application, and its evolution towards SMARTA 3, demonstrates strong potential to contribute to **safer, more inclusive, and sustainable urban mobility systems**. By embedding user feedback and stakeholder input into its continuous development, the Municipality of Trikala is well-positioned to deliver a **scalable and replicable smart mobility solution** within the European context.

Annexes

Annex I – SMARTA Application User Survey Questionnaire (Trikala Pilot)

A.1 Introduction

This questionnaire was developed in the framework of the **ELABORATOR project** to assess user perceptions of the SMARTA mobility application implemented in Trikala.

The aim of the survey is to evaluate key aspects of the application, including usability, functionality, mobility services, and inclusiveness, in order to support its continuous improvement and future deployment (SMARTA 3).

All responses are anonymous and were collected using a **five-point Likert scale (1–5)**, where:

- **1 = Strongly Disagree**
- **2 = Disagree**
- **3 = Neutral**
- **4 = Agree**
- **5 = Strongly Agree**

A.2 Questionnaire Structure

The questionnaire is structured into four thematic sections:

- **Section A:** Usability
- **Section B:** Functionality & Mobility Services
- **Section C:** Inclusiveness & Accessibility
- **Section D:** Overall Evaluation
- **Section E:** Open Feedback (optional)

A.3 Survey Questions

Section A: Usability

No.	Question	Scale (1–5)
Q1	The application is well organized and user-friendly	☐1 ☐2 ☐3 ☐4 ☐5
Q2	Navigation within the application is simple and understandable	☐1 ☐2 ☐3 ☐4 ☐5
Q3	The information provided is clear and informative	☐1 ☐2 ☐3 ☐4 ☐5

Section B: Functionality & Mobility Services

No.	Question	Scale (1–5)
Q4	The application meets my mobility needs in Trikala	☐1 ☐2 ☐3 ☐4 ☐5
Q5	The application supports the use of sustainable mobility infrastructure	☐1 ☐2 ☐3 ☐4 ☐5
Q6	The application provides reliable information about bus routes and stops	☐1 ☐2 ☐3 ☐4 ☐5
Q7	The application helps me locate shared bike stations	☐1 ☐2 ☐3 ☐4 ☐5
Q8	The application helps me locate parking spaces	☐1 ☐2 ☐3 ☐4 ☐5
Q9	The application performs well (speed and technical reliability)	☐1 ☐2 ☐3 ☐4 ☐5
Q10	The application is reliable in real-life conditions	☐1 ☐2 ☐3 ☐4 ☐5

Section C: Inclusiveness & Accessibility

No.	Question	Scale (1–5)
Q11	The application is accessible for vulnerable groups (e.g. elderly, people with disabilities)	☐1 ☐2 ☐3 ☐4 ☐5
Q12	The application covers the mobility needs of vulnerable groups	☐1 ☐2 ☐3 ☐4 ☐5
Q13	The application is accessible for migrants (e.g. language support)	☐1 ☐2 ☐3 ☐4 ☐5
Q14	The application considers the mobility needs of women	☐1 ☐2 ☐3 ☐4 ☐5

Section D: Overall Evaluation

No.	Question	Scale (1–5)
Q15	Overall, I am satisfied with the application	☐1 ☐2 ☐3 ☐4 ☐5
Q16	I would recommend this application to others	☐1 ☐2 ☐3 ☐4 ☐5

Section E: Open Feedback (Optional)

Q17. What do you like most about the application?

....

Q18. What improvements would you suggest?

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